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Bihar Engineering University, Patna

B.Tech 5th Semester Exam., 2021

Time: 3 hours

Code: 101502 ***Subject: Environmental Engineering-I*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.
(v) Assume data suitably, if not given.

Q.1 Choose the correct answer of the following (any seven): **[14]**

(a) The suitable layout of a distribution system for irregularly growing town is
(i) dead-end system (iii) radial system
(ii) grid-iron system (iv) ring system

(b) Average rate of water consumption per head per day as per Indian Standard is
(i) 100 litres (iii) 165 litres
(ii) 135 litres (iv) 200 litres

(c) Orthotolidine test is used for the determination of
(i) dissolved oxygen (iii) biochemical oxygen demand
(ii) residual chlorine (iv) dose of coagulant

(d) Colour of water is expressed in number of a
(i) po value (iii) platinum cobalt scale
(ii) silica scale (iv) None of the above

(e) MPN index is a measure of which one of the following?
(i) Coliform bacteria (iii) Dissolved oxygen content
(ii) BOD_5 (iv) Hardness

(f) When environmental lapse rate is more than adiabatic lapse rate, then the environment is said to be
(i) stable (iii) neutral
(ii) unstable (iv) None of the above

(g) The reference pressure used in determination of sound pressure level is
(i) 20 dB (iii) 20 μPa
(ii) 10 μPa (iv) 10 dB

(h) The frequency of sound is measured in
(i) hertz (iii) decibel (dB)
(ii) Dobson unit (DU) (iv) None of the above

(i) The most hazardous gaseous air pollutant for human health is
(i) nitrogen (iii) oxygen
(ii) carbon dioxide (iv) sulphur dioxide

(j) Among the given choices below, which of the following is not a primary air pollutant?
(i) PAN (iii) Suspended particulate matter
(ii) Volatile organic compounds like hydrocarbon (iv) Oxides of nitrogen

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Q.2 Solve both questions : **[7+7=14]**

(a) Explain per capita demand of water. How is it determined? Discuss various factors that affect the rate of demand.

(b) Explain the importance of chemical and bacteriological analysis of water used for domestic purpose.

Q.3 Solve both questions : **[7+7=14]**

(a) Classify the atmosphere on the basis of temperature variation. State main activities taking place in each of them.

(b) What is BOD? Deduce the expression for first-stage BOD.

Q.4 Solve both questions : **[7+7=14]**

(a) Explain disinfection process. List the various methods used for disinfection process.

(b) Differentiate between temporary and permanent hardness.

Q.5 Solve both questions : **[7+7=14]**

(a) Enumerate the different major air pollutants, their characteristics, sources and health effects on human beings in a tabular form.

(b) What do you mean by gaseous pollutants? Name and describe any one controlling device used for gaseous pollutants.

Q.6 Solve this question : **[14]**

The flow through a flocculator processes 300 m^3/hr of water at 20°C. The paddles are arranged longitudinally. The optimum G value has been found from jar tests to be 600 s^{-1} and detention time of 30 s. Determine (a) basin dimensions, (b) power applied to the water and (c) paddle configuration and rotational speed. Assume any other data not given.

Q.7 Solve this question : **[14]**

(a) Write down the Indian Standard of ambient noise level.

(b) Write on the sources, effects and control measures of noise pollution.

Q.8 Solve both questions : **[7+7=14]**

(a) How would you assess the water demand of the building to be used for different purposes?

(b) How is the hot water requirement of building computed?

Q.9 Write short notes on (any four): **[14]**

(a) Ion-exchange method of water treatment
(b) Head loss in screening
(c) Inverted syphon
(d) Break-point chlorination
(e) Cyclone collectors

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